CALL FOR PAPERS - *IEEE Transactions on Aerospace and Electronic Systems*
Special Section on *Low-Altitude Wireless Networks for Resilient Aerospace Operations*

Low-Altitude Wireless Networks (LAWNs), which interconnect UAVs, drone swarms, eVTOL aircraft, ground infrastructure, sensing assets, and edge processors, are emerging as a new class of aerospace electronic systems. Beyond providing airborne connectivity, LAWNs enable cooperative sensing, airspace situational awareness, positioning and navigation, resilient command and control, remote identification, and autonomous mission coordination in dynamic 3D airspace.

Operation in low-altitude environments introduces distinctive challenges, including rapid 3D mobility, air-ground and air-air propagation, Doppler effects, clutter, interference, GNSS degradation, jamming and spoofing risks, SWaP limitations, and stringent safety requirements. These challenges call for integrated methods spanning radar and RF sensing, multisensor fusion, PNT, communication and waveform design, C2, autonomous control, AI-enabled adaptation, and field validation. This Special Section seeks original contributions that advance the theory, methods, systems, and experiments of LAWNs from an aerospace electronic systems perspective. Topics of interest include, but are not limited to:

- RF/radar sensing, passive/active sensing, multifunction sensing, and cognitive radar for LAWNs
- Target detection, tracking, classification, identification, and low-altitude airspace surveillance
- Radar resource management, sensor scheduling, and adaptive sensing for dynamic 3D airspace
- Multisensor fusion using RF, radar, EO/IR, LiDAR, inertial, acoustic, and environmental sensors
- Cooperative localization, relative positioning, navigation, and PNT integrity for UAVs, eVTOL aircraft, drone swarms, and ground infrastructure
- GNSS-denied, GNSS-degraded, and spoofing-resilient PNT, including uncertainty quantification
- Network-aided positioning, radar-aided navigation, remote identification, tracking, and surveillance
- Waveform design for joint sensing, positioning, communication, and command-and-control in LAWNs
- Doppler-resilient transmission, estimation, synchronization, and tracking for high-mobility LAWNs
- 3D array processing, adaptive beamforming, altitude-aware precoding, and MIMO/mmWave/THz sensing and communications
- Resilient C2, telemetry, spectrum sharing, interference management, and coexistence for safety-critical low-altitude missions
- Autonomous guidance, path planning, swarm coordination, and air-ground cooperative control enabled by networked sensing and positioning
- AI-enhanced estimation, detection, beam management, adaptive sensing, cooperative localization, anomaly detection, edge intelligence, digital twins, and multi-agent coordination
- Field trials, hardware prototypes, datasets, experimental platforms, validation methodologies, standardization, interoperability, certification, and regulatory frameworks for deployable LAWNs

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